

Pharmacology of General Anesthetics Part 1

Inhaled anesthetics, General Principles

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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After completing the preparation materials, students should be able to:

Inhaled anesthetics

- Identify the proposed mechanisms of action of the inhaled anesthetics and relate them to the desired effects.
- Explain speed of onset and rate of recovery in relation to the F_A/F_I ratio for inhaled anesthetics.
- Describe how solubility in blood, alveolar blood flow, and the difference in partial pressure between the alveolar blood gas and venous blood affect uptake of inhaled anesthetic.
- Apply the minimum alveolar concentration (MAC) to its clinical application in anesthesia.
- Differentiate the properties and the side effects of the inhaled anesthetics in common use.

Intravenous anesthetics

- Identify the proposed mechanisms of action of the intravenous anesthetics and relate them to the desired effects.
- Explain redistribution and context-sensitive half-time in relation to recovery from a single dose and from repeated bolus doses of continuous infusion of I.V. anesthetics.
- Differentiate the specific intravenous general anesthetics in relation to their clinical uses and their drug-specific side effect profiles.
- Define the goal of balanced anesthesia.
- Explain the typical method for achieving balanced anesthesia.

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions | See next slide.

Optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- *Use the resources that work best for you.*
- *You do not need to study all of them.*
- *Work through the GUIDED READING QUESTIONS with pen/pencil and paper.*

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

- *Practice questions (not graded): Simple Recall and Case Vignettes*

Links to resources

ScholarRx Brick: Neurology and Special Senses > Disorders of the Nervous System and Special Senses > Pain Mechanisms and Treatment > Diagnostic Approach to Disorders of the Nervous System and Special Senses

Principals of Anesthesia > General Anesthetics <https://exchange.scholarrx.com/brick/general-anesthetics>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024: Chapter 25: General Anesthetics

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3382§ionid=281750826>

Access Medicine Katzung's Pharmacology Examination & Board Review, 14e, 2024; Chapter 25: General Anesthetics

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3461§ionid=285593753>

LWW Health Libraries, Premium Basic Sciences: Lippincott's Illustrated Reviews, 8e, 2023; Chapter 13: Anesthetics

<https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253327491&bookid=3222>

Anesthesia Textbook Chapters:

Clinical Key: Anesthesia Secrets, 6e, 2021; Chapter 14: Inhaled Anesthetics

<https://www-clinicalkey-com.nyit.idm.oclc.org/#!/content/book/3-s2.0-B9780323640152000147>

Clinical Key: Anesthesia Secrets, 6e, 2021: Chapter 15: Intravenous Anesthetics

<https://www-clinicalkey-com.nyit.idm.oclc.org/#!/content/book/3-s2.0-B9780323640152000159>

Access Medicine: Morgan & Mikhail's Clinical Anesthesia, 7e, 2022; Section II: Clinical Pharmacology, Chapter 8: Inhalational Anesthetics

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3194§ionid=266517982>

Access Medicine: Morgan & Mikhail's Clinical Anesthesia, 7e, 2022; Section II: Clinical Pharmacology, Chapter 9: Intravenous Anesthetics

<https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3194§ionid=266518173>

Terminology

General anesthesia	Medically induced state of unconsciousness accompanied by a loss of protective reflexes, resulting from the administration of one or more anesthetic agents.
Balanced anesthesia	An approach to anesthesia that uses a combination of different drugs and techniques to achieve the desired effects of general anesthesia while minimizing side effects
Volatile anesthetic	A type of inhaled anesthetic that is in a liquid form at room temperature and delivered to the patient via a vaporizer connected to an anesthesia machine
Partition coefficient (λ) (anesthesia)	The relative affinity of an anesthetic agent for two different phases, such as blood and gas or oil and gas, and how that anaesthetic distributes itself between the two phases when equilibrium has been achieved
F_A	The concentration of anesthetic gas in alveoli
F_I	The concentration of anesthetic gas in inspired air
F_A/F_I ratio	Ratio of the alveolar gas concentration to the inspired gas concentration F_A/F_I ratio indicates how quickly the concentration of the anesthetic in the alveoli approaches the concentration of gas in the inspired air. F_A/F_I ratio is a key concept in the pharmacokinetics of inhaled anesthetic gases.

Terminology

MAC	The alveolar concentration that prevents movement in 50% of patients in response to a standardized stimulus (eg, surgical incision).
Redistribution	The process by which a highly lipid soluble intravenous drug moves from the central compartment to peripheral tissues (such as muscle and fat) after initial administration, reduces the concentration of the drug in the central compartment, leading to a decrease in its anesthetic effect. The termination of the anesthetic effect after a single bolus dose is primarily due to redistribution rather than metabolism or excretion. As the drug moves away from the brain to other tissues, its concentration in the brain decreases, leading to the patient regaining consciousness.
Context-sensitive half-time	The time required for the plasma concentration of a drug to decrease by 50% after discontinuing an infusion that has been running for a specific duration. The "context" refers to the duration of the infusion. The context-sensitive half-time varies depending on how long the drug infusion has been administered. Longer infusions typically result in longer context-sensitive half-times due to the accumulation of the drug in peripheral tissues

General Anesthetics Introduction

- Sedation of patients facilitates surgical interventions provides reduced anxiety, anterograde amnesia, skeletal muscle relaxation, suppression of reflexes, analgesia, and general anesthesia. No single agent provides all of these objectives. Several categories of drugs are combined to produced optimal sedation.
- Preoperative anxiolytics and analgesics mitigate the anxiety and pain of the procedure.
- Neuromuscular blockers enable endotracheal intubation and muscle relaxation to facilitate surgery.
- Inhaled or intravenous general anesthetics are used to rapidly induce and/or maintain a state of general anesthesia.
- Inhaled gases are used primarily for maintenance of anesthesia after administration of an intravenous anesthetic. Depth of anesthesia can be rapidly altered by changing the inhaled gas concentration. Inhalational anesthetics are delivered in a recirculation system that a CO₂ absorber to remove carbon dioxide, preventing CO₂ accumulation and allow rebreathing of the gas.
- Inhalational anesthetics have steep dose-response curves with very narrow therapeutic windows (small therapeutic indices). The difference in concentrations that elicit general anesthesia and the concentrations that induce cardiopulmonary collapse is small. There are no antagonists that reverse general anesthetics.



General Anesthetics Introduction

- **Goals** for induction of anesthesia are to rapidly and pleasantly produce hypnosis, amnesia, analgesia, akinesia, and autonomic and sensory block while maintaining oxygenation, ventilation, and hemodynamic stability.
- The goals for the maintenance phase of anesthesia are to maintain Stage III anesthesia at a safe anesthetic depth and while maintaining respiratory and hemodynamic stability.
- **Intravenous anesthetics** cause rapid induction of anesthesia, often occurring in a minute or less. It is the most frequent method for inducing anesthesia before the maintenance phase with an inhaled agent. IV anesthetics may be used as single agents for short procedures or administered as infusions for maintenance anesthesia during longer surgeries (total intravenous anesthesia).
- **Inhaled anesthetics** in common use are the volatile halogenated agents, isoflurane, desflurane, and sevoflurane, and nitrous oxide, which is a gas at standard temperature and ambient pressure.
- Inhaled anesthetics are taken up by the pulmonary circulation through gas exchange in the alveoli and distribute to the CNS and other body compartments.
- **When the partial pressure of the anesthetic gas in the inspired air, alveoli, arterial blood, and CNS are in equilibrium (steady state), the desired clinical effect will be produced.**
- **Equilibrium ensures that the anesthetic concentration in the CNS is stable and effective for maintaining anesthesia during the procedure.**

Inhaled Anesthetics Key Points

- **The F_A/F_I ratio is the key determinant of the speed of inhaled anesthesia induction and recovery.**
 - The fractional rise of anesthetic partial pressure during induction is expressed as a ratio of alveolar concentration (F_A) over inspired concentration (F_I).
 - The faster F_A/F_I approaches 1 (representing inspired-to-alveolar equilibrium), the faster anesthesia onset will be during inhaled induction of anesthesia.
- F_A/F_I ratio is influenced by increasing the delivered concentration of anesthetic, alveolar ventilation, the solubility of the anesthetic (defined by the blood:gas partition coefficient), cardiac output, alveolar to venous partial pressure difference, and the volume of the breathing circuit and fresh gas flow.
- Rate of recovery when the anesthetic is withdrawn is the reverse of the principles that influence induction.
- **Potency** is quantitatively defined as the minimum alveolar concentration (MAC) that produces immobility in 50% of patients exposed to a standardized stimulus, such as pain of a surgical incision. It is used as a measure of anesthetic in the alveoli and, thus, the concentration in the central nervous system.
- MAC is the median effective dose (ED_{50}) of the anesthetic, expressed as the percentage of gas in a mixture required to achieve the desired anesthetic effect. Quantitatively in vol%, MAC is small for potent anesthetics (eg isoflurane) and large for less potent anesthetics (eg nitrous oxide).
- MAC is decreased by old age, hyponatremia, hypothermia, opioids, barbiturates, alpha-2 adrenergic agonists, acute alcohol intoxication, and pregnancy. MAC is increased by young age (6-12 months), hyperthermia, hypernatremia, chronic alcoholism, and CNS stimulants.

Inhaled Anesthetics Key Points

- **Mechanisms of action** likely include effects on ion channels by interactions with membrane lipids or proteins and subsequent effects on central neurotransmitters: 1) facilitated GABA-mediated inhibition, 2) block of central excitatory NMDA glutamate receptors and acetylcholine nicotinic receptors. Nitrous oxide is a noncompetitive NMDA receptor antagonist.
- **Muscle effects:** All potent (volatile) inhaled anesthetics induce dose-dependent relaxation of skeletal and smooth muscle through several mechanisms.
- **CNS effects:** Inhaled anesthetics decrease the metabolic activity of the brain, cause dose-dependent cerebral vasodilation and increase intracranial pressure, and blunt cerebral autoregulation.
- **Cardiovascular effects:** Isoflurane, desflurane, and sevoflurane primarily cause vasodilation with relatively little initial effect on myocardial force of contraction (inotropy), which preserves cardiac output. Isoflurane and desflurane tend to produce dose-dependent tachycardia because they cause less depression of the baroreflex. Desflurane can cause transient activation of the sympathetic nervous system and elevated catecholamine levels, resulting in tachycardia and hypertension. The older agents halothane and enflurane (no longer in clinical use in the US) cause dose-dependent myocardial depression resulting in reduced cardiac output and hypotension.
- **Respiratory effects:** Airway irritation by isoflurane and desflurane may cause coughing, laryngospasm, or bronchospasm. Sevoflurane and nitrous oxide are nonpungent. Sevoflurane's bronchodilator effects make it a preferred agent for patients with underlying breathing problems. Dose-dependent respiratory effects (Stage III/IV) blunt ventilatory response to hypercapnia and hypoxic ventilatory drive. Ventilatory assist becomes necessary depending on the stage of anesthetic depth and the patient's clinical status¹¹.

Inhaled Anesthetics Key Points

- **Desiccated CO₂ absorbers** in breathing circuits react with the halogenated anesthetics passing through them, causing production of carbon monoxide (CO):
 - Relative likelihood of CO production: desflurane > isoflurane > > sevoflurane.
- **Nitrous oxide** is a weak anesthetic – it does not achieve full anesthesia when used alone. It is a noncompetitive NMDA receptor antagonist with significant analgesic and sedative properties. It is mainly used as an adjunct to other anesthetics.
- N₂O causes body space distension, diffusion hypoxia, second gas effect, and inactivates vitamin B12. N₂O should be avoided in patients with increased middle ear pressure, bowel distention, pneumothorax, pneumoperitoneum, or pneumocephalus and those with megaloblastic anemia (vitamin B12 deficiency). Diffusion hypoxia occurs on discontinuation; O₂ rather than air should be administered when N₂O is discontinued.
- **N/V:** All inhaled anesthetics increase the risk of post-operative nausea and vomiting.
- **Environment:** All inhaled anesthetics, including nitrous oxide, are greenhouse gases.

Intravenous Anesthetics Key Points

- Currently used general anesthetics do not provide the five desired effects of an ideal anesthetic: unconsciousness, amnesia, analgesia, inhibition of autonomic reflexes, and skeletal muscle relaxation.
- **Balanced anesthesia technique** employs multiple drugs in smaller doses to achieve the desired combination of effects while minimizing unwanted effects by using less of each drug than if it were administered alone.
- **Redistribution** describes the process in which a single bolus of a lipophilic drug is initially distributed to highly perfused organs (eg the brain, liver, kidneys) and then moves to tissues with lower blood flow (eg muscle, fat), resulting in a short duration of effect.
- **Context-sensitive half-time** describes the elimination half-time after a continuous intravenous infusion of a drug as a function of the duration of the infusion. Initial anesthetic effects may be terminated by redistribution from CNS tissues; full recovery occurs when the drug is eliminated from the body by metabolism and excretion.
- Some drugs have short, stable context-sensitive half-times and effects wear off quickly and predictably after stopping the infusion while others have longer context-sensitive half-times and take longer to wear off and are less predictable.
- Appropriate dosing of intravenous anesthetics requires consideration of intravascular volume status, comorbidities, age, and chronic medications.

Intravenous Anesthetics Key Points

Propofol, ketamine, etomidate, midazolam, opioids (primarily fentanyl and remifentanyl), and dexmedetomidine, and lidocaine are the intravenous anesthetics used most frequently in contemporary anesthetic practice.

- **Propofol** is a GABA-A receptor positive allosteric modulator widely used for induction and maintenance of anesthesia and in monitored anesthesia care.
 - Propofol has antiemetic and anticonvulsant properties but does not provide analgesia. It can cause respiratory depression and is associated with bradyarrhythmias and ECG abnormalities. Injection is painful.
 - Propofol is formulated with soybean oil and egg lecithin and supports bacterial growth. It should be avoided in children with known anaphylaxis to eggs.
- **Etomidate** is an ultra-rapid acting GABA-A receptor positive allosteric modulator used for induction and maintenance of anesthesia.
 - Etomidate has negligible cardiac and vascular effects. It may be a good option for patients with cardiovascular instability or who are critically ill because it. It does not provide analgesia.
 - Etomidate inhibits 11-beta-hydroxylase, which is involved in steroidogenesis, and may transiently decrease cortisol plasma levels. Therefore, it should not be infused for extended periods.
 - Etomidate causes myoclonus on induction, which is not related to convulsions.
 - Injection site pain, nausea and vomiting, and involuntary skeletal movements are common.

Adjunctive Intravenous Agents Key Points

- **Ketamine** is a short-acting NMDA receptor inhibitor used as an adjunct in surgical anesthesia and monitored anesthesia care. It produces a dissociative state with profound analgesia.
 - Ketamine has excellent bronchodilator properties and maintains airway reflexes respiratory drive.
 - Ketamine stimulates sympathetic outflow with increases in blood pressure, heart rate, cardiac output, pulmonary artery pressure, and intracranial pressure.
 - Psychomimetic effects may cause hallucinations, nightmares, and vivid dreams, which limit its use.
 - Ketamine is typically used as an adjunctive agent for patients undergoing painful surgery to limit opioid use.
- **Benzodiazepines**, frequently midazolam, are GABA-A receptor positive allosteric modulators that provide sedation and anterograde amnesia in anesthetic techniques.
 - Benzodiazepines have minimal cardio-depressant effects.
 - Benzodiazepines can cause respiratory depression, especially when combined with other CNS depressants, such as opioids.

Adjunctive Intravenous Anesthetics Key Points

- **Dexmedetomidine** is an alpha-2 adrenergic receptor agonist. It has sedative, analgesic, anxiolytic, and sympatholytic properties.
 - It is used in intensive care units, surgery and monitored anesthetic care.
 - It does not cause respiratory depression and blunts undesirable cardiovascular responses.
 - It can cause bradycardia and hypotension and increased time to emergence.
- **Opioids**, frequently fentanyl or remifentanyl, rapidly provide analgesia and have a short duration of effect.
 - Opioids do not have amnestic properties.
 - Opioids cause respiratory depression, hypotension, and postoperative nausea and vomiting.
- **Lidocaine IV infusion** may be used during induction of general anesthesia to suppress the cough reflex and laryngospasm, reduce the pain of injection or other induction agents, and supplement anesthetic effects.
 - Lidocaine-induced ventricular rate can contribute to arrhythmia. Arrhythmias are uncommon when IV lidocaine is used for pain control.

Operating room



<http://www.oasor.com/patient-info/what-is-anesthesia>

General anesthesia

Balanced anesthesia

- **General anesthesia** induces a physiological state that typically includes analgesia, amnesia, loss of consciousness, inhibition of sensory and autonomic reflexes, and skeletal muscle relaxation
- Currently used general anesthetics do not provide the five desired effects of an ideal anesthetic: unconsciousness, amnesia, analgesia, inhibition of autonomic reflexes, and skeletal muscle relaxation.
- **Balanced anesthesia technique:** Modern anesthesia typically involves a combination of intravenous and inhaled drugs – neuronal depressants.
 - Additive/synergistic favorable effects allowing lower doses of multiple agents, which minimizes their adverse effects
 - Minimizes patient risk and maximizes patient comfort and safety.
 - Examples: preoperative benzodiazepines, induction of anesthesia with an intravenous anesthetic, and maintenance of anesthesia with a combination of inhaled and intravenous drugs or total intravenous anesthesia, plus opioid for pain, and neuromuscular blocker to facilitate tracheal intubation and surgery.
- Other anesthesia techniques include total intravenous anesthesia and monitored anesthesia care.



Types of General Anesthetics

Inhaled anesthetics

- Volatile anesthetics: Liquids at room temperature and sea level ambient pressure
- Gases: Gases at typical ambient temperatures and pressures

Intravenous anesthetics

- Lipid soluble drugs that preferentially partition into highly perfused lipophilic tissues (brain, spinal cord) → rapid onset of action
- Redistribution → short duration of action (single dose)
- Context-dependent half-times (infusion and repeated doses)



Phases of General Anesthesia

Induction phase:

The time from the administration of the anesthetic to the onset of effective surgical anesthesia in the patient

The induction phase is important for avoiding the dangerous excitatory stage, characterized by delirium and irregular respiration.

Maintenance phase:

Sustained surgical anesthesia

Recovery phase:

The time from withdrawal of anesthetic postoperatively to return to consciousness

It depends on the rate of removal of the drug from the brain.

The patient is monitored for return to consciousness and for delayed toxic reactions.

Stages of anesthetic depth

Stage I	Analgesia state: Patient is conscious and rational, with decreased perception of pain.
Stage II	Delirium stage: Patient is unconscious; body responds reflexively; irregular breathing pattern with breathholding.
Stage III	Surgical anesthesia: Increasing degrees of muscle relaxation; unable to protect airway.
Stage IV	Medullary depression: There is depression of cardiovascular and respiratory centers.

Adapted from: Hewer CL. The stages and signs of general anesthesia. Br Med J 1937; 2:274.

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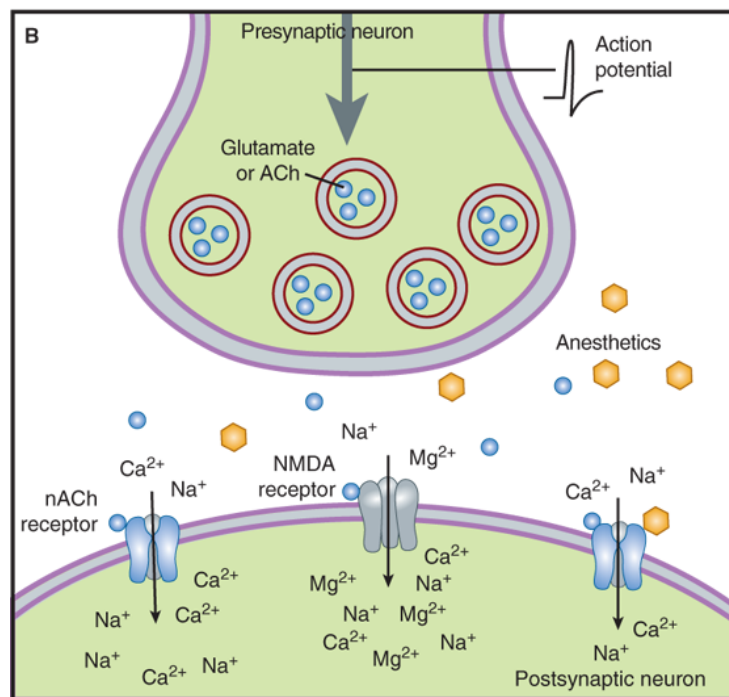
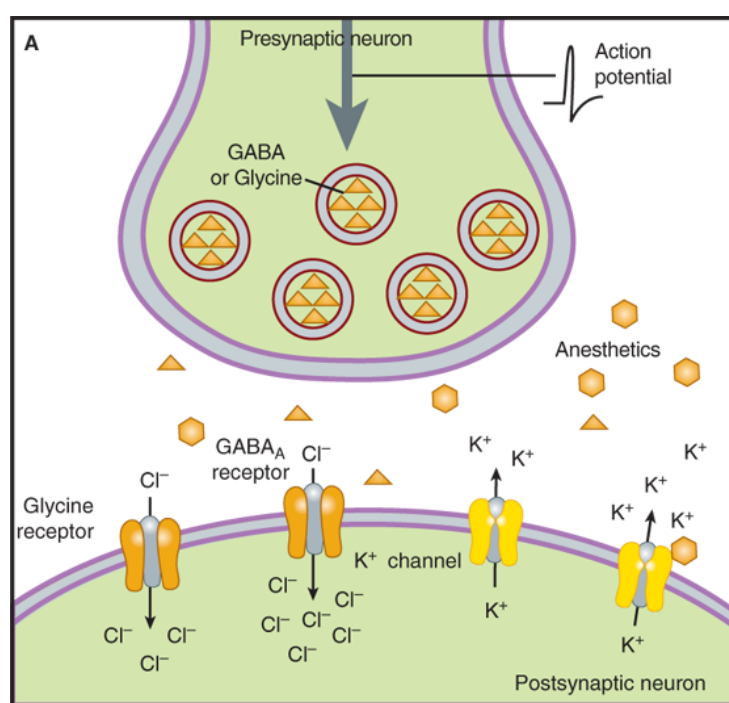
Signs indicating stages of anesthesia

	Respiration		Ocular movements	Pupils no premedication	Eye reflexes	Pharynx larynx reflexes	Lacrimation	Muscle tone	Respiratory response incision	
	Intercostal	Diaphragm								
Stage I			Voluntary control				Normal	Normal		
Stage II						Swallow				
Stage III Plane 1					Lid tone	Retch				
						Vomit				
Plane 2						Glottis				
Plane 3					Pupillary light reflex					
Plane 4						Carinal				
Stage IV										

Modified from: Gillespie NA. The signs of anesthesia. Anesth Analg 1943; 22:275. Copyright © 1943 International Anesthesia Research Society. Reproduced with permission from Wolters Kluwer Health. Unauthorized reproduction of this material is prohibited.

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- **The goals for induction of anesthesia** are to rapidly and pleasantly produce hypnosis, amnesia, analgesia, akinesia, and autonomic and sensory block while maintaining oxygenation, ventilation, and hemodynamic stability
- **The goals for the maintenance phase of anesthesia** are to maintain Stage III anesthesia at a safe anesthetic depth and while maintaining respiratory and hemodynamic stability.



Proposed Mechanisms of Action:

Anesthetics affect neurons at various cellular locations.

Anesthetic drugs may...

Panel A: Enhance inhibitory synaptic activity, or

Panel B: Diminish excitatory activity

Actions at different locations may be cumulative.

Nitrous oxide:
Noncompetitive
NMDA
antagonist

Molecular targets that have been studied

Inhibitory

- GABA-gated chloride channels
- Glycine-gated chloride channels
- Potassium channels

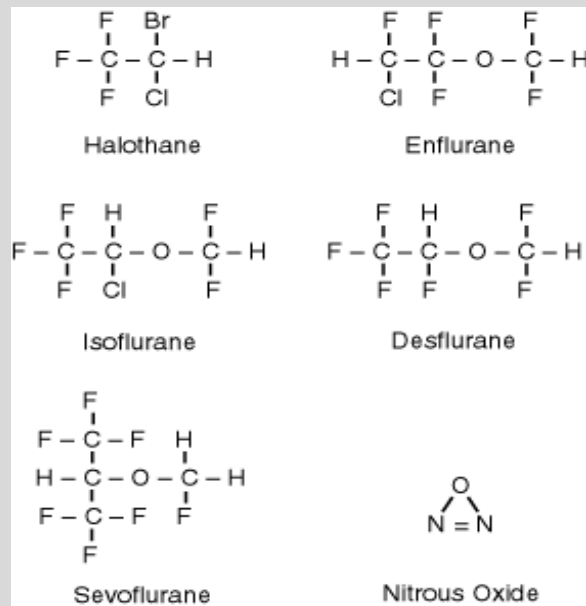
Excitatory

- Ion channels activated by ACh (nicotinic and muscarinic)
- Glutamate AMPA, kainite, NMDA receptors
- Serotonin 5-HT_{2/3}

- Neural networks altered by general anesthetics

ACh: acetylcholine, GABA: γ-aminobutyric acid.

Source: Todd W. Vanderah:
Basic & Clinical Pharmacology, Sixteenth Edition
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Inhaled Anesthetics

In common use

Not in common use

Volatile anesthetics

(liquids at room temperature and vaporized for inhalation)

*Isoflurane

*Halothane

*Desflurane

Enflurane

*Sevoflurane

Gases

(gases at room temperature and standard pressures)

*Nitrous Oxide

Xenon (noble gas)



photo: Richard Struthers,
The Anesthesia Machine,
<http://www.anes.med.umich.edu/machine/>

Pharmacological Properties of Inhaled Anesthetics

Halogenated anesthetics

Anesthetic	B:G	B:B	MAC	Metabolism	Comments	Pungency
Nitrous oxide	0.47	1.1	> 100	None	Incomplete anesthetic , rapid onset and recovery	Nonpungent Suitable for induction
Desflurane	0.42	1.3	6–7	< 0.05%	Rapid onset and recovery	Most Pungent, high incidence airway irritation → coughing, bronchospasm
Sevoflurane	0.69	1.7	2.0	2–5% (fluoride)	Rapid onset and recovery unstable in soda-lime, forms Compound A	Nonpungent, bronchodilation action; preferred for pts. with airway problems
Isoflurane	1.40	2.6	1.40	< 2%	Medium rate of onset and recovery	Pungent, airway irritation → coughing, bronchospasm
Enflurane	1.80	1.4	1.7	8%	Medium rate of onset and recovery	Pungent
Halothane	2.30	2.9	0.75	> 40%	Medium rate of onset and recovery	Nonpungent

B:G=blood:gas partition coefficient; B:B=brain:blood partition coefficient

Partition coefficient: The ratio of anesthetic gas concentration partitioning between two tissues at equilibrium (at 37°C).

MAC is the anesthetic concentration that produces immobility in 50% of patients exposed to a noxious stimulus.

Katzung, 2024; Table 25-1



Inhaled Anesthetics

Uptake, Distribution, and Elimination

Key pharmacokinetic concept: Equilibrium

When the partial pressure of the anesthetic gas in the inspired air, alveoli, arterial blood, and CNS are in equilibrium (steady state), the desired clinical effect will be produced.

Equilibrium ensures that the anesthetic concentration in the CNS is stable and effective for maintaining anesthesia during the procedure.



Pharmacokinetic Principles of General Anesthesia

Anesthetic uptake into the blood is determined by pharmacokinetic characteristics of each anesthetic agent as well as patient factors.

- Gases move from one body compartment to another according to the partial pressure gradients.
- Inhaled anesthetics are taken up through gas exchange between the alveoli of the lung and the pulmonary circulation for distribution to the body compartments.
- Four factors influencing uptake of anesthetic between the lung and the blood (and thus the speed of anesthetic induction) are the:
 - Inspired concentration of anesthetic gas
 - Solubility of gas in blood (defined by the blood:gas partition coefficient)
 - Cardiac output
 - Alveolar-venous partial pressure difference

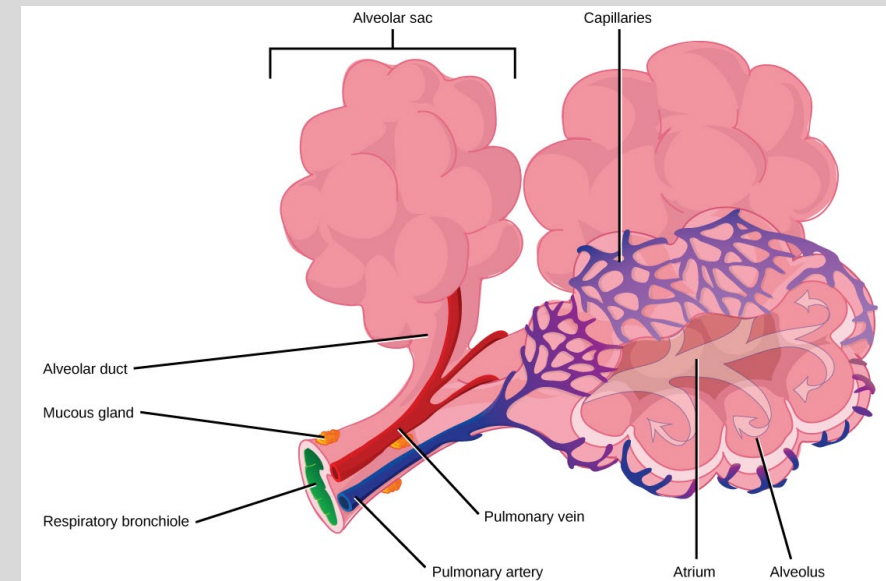


Figure: BC Campus Open Education
<https://opentextbc.ca/biology/chapter/20-1-systems-of-gas-exchange/>

The speed of anesthetic exchange between the lungs and blood is a crucial factor in determining the speed of onset of anesthesia.

Inspired Concentration and Uptake of Inhaled Anesthetics

F_A/F_I ratio: The **driving force** for gas uptake is the **ratio** between **inspired** concentration (partial pressure) and **alveolar** concentration.

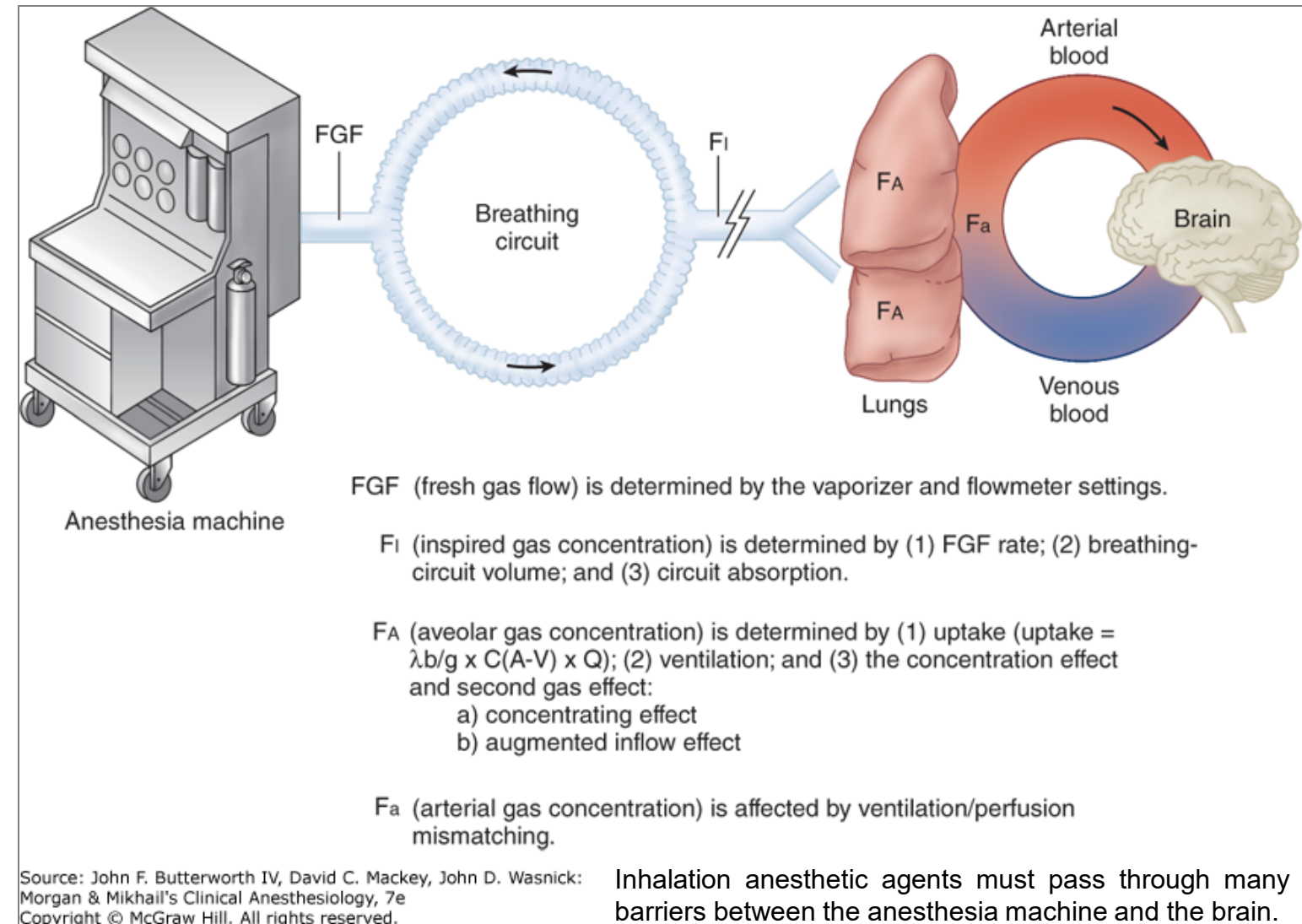
Equilibrium ($F_A/F_I=1$) is achieved when the partial pressure in the inspired gas is equal to the partial pressure in the alveolar gas.

F_A : concentration of alveolar gas

F_I : concentration of inspired gas

Alveolar partial pressure determines how much anesthetic gas will dissolve in the blood – the higher the partial pressure in the alveolus, the higher the partial pressure in the blood.

When steady state has been reached,
the partial pressures of the gas in inspired air $\approx$ alveolar air $\approx$ arterial blood $\approx$ brain $\rightarrow$ clinical effect.



The faster the uptake of anesthetic agent, the greater the difference between inspired and alveolar concentrations and the slower the rate of induction.

Why?

- Alveolar concentrations lag behind inspired concentrations ($F_A < F_I$) because anesthetics are taken up by the pulmonary circulation during induction.

High uptake

- When uptake is fast, a large amount of anesthetic is absorbed into the blood from the alveoli.

Concentration difference

- Rapid uptake creates a significant difference between the inspired concentration and the alveolar concentration ($F_A < F_I$) as the alveoli are depleted of anesthetic faster than they can be replenished by inhalation.

Speed of induction

- Anesthesia induction depends on equilibrium between the alveolar and CNS concentrations. When uptake is high, it takes longer to reach steady state ($F_A/F_I = 1$).

Ventilation

Increasing the alveolar ventilation continuously replaces anesthetic taken up by the pulmonary bloodstream → better maintenance of alveolar concentration

- By increasing ventilation, the anesthesiologist can increase the rate of induction.
- Increasing the rate and depth of breathing (ventilation) delivers more fresh gas to the alveoli per unit of time.

Minute ventilation in mL/min = tidal volume x respiratory rate

- Increasing the partial pressure of the anesthetic in the alveoli creates a steeper gradient, which leads to faster equilibration with arterial blood.
- Thus, the partial pressure of the anesthetic in the brain required to produce the clinical effect is achieved more quickly, which increases the speed of induction.



Why is the speed of induction slower with the more soluble anesthetic gases?

more soluble → greater capacity for uptake → slower rate of rise in partial pressure → longer time for F_A/F_I to approach equilibrium

less soluble → blood is quickly saturated → rapid rate of rise in partial pressure → equilibrium is quickly established

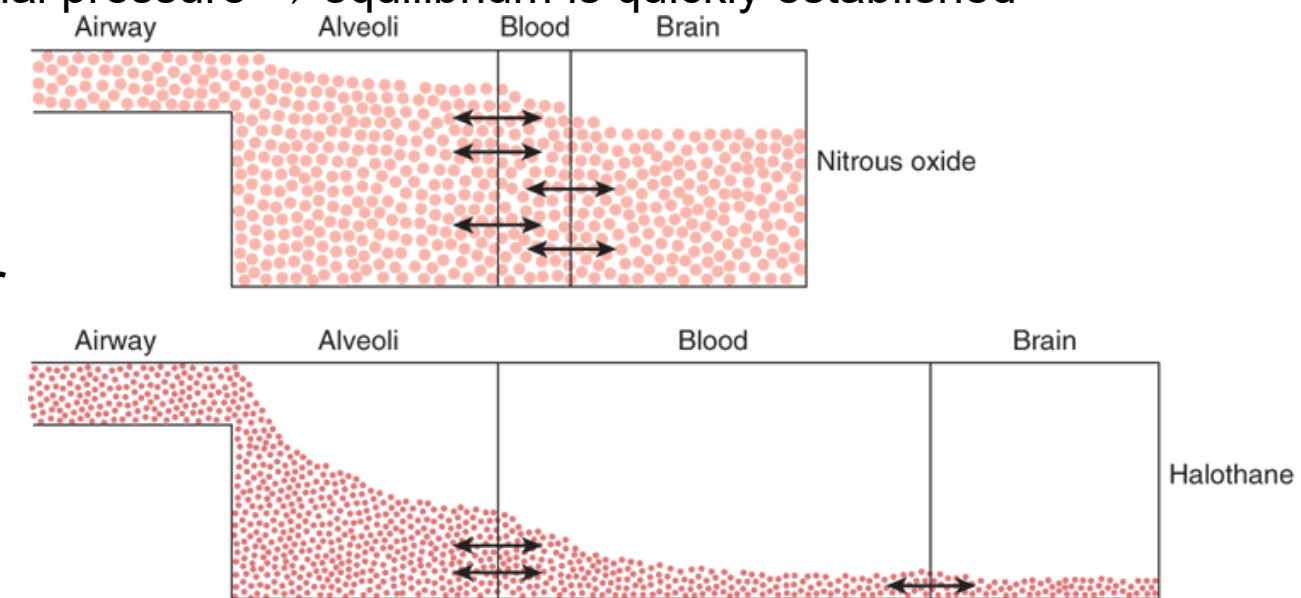
An inhaled anesthetic's blood:gas partition coefficient defines the relative affinity of a gas for the blood compared to its affinity for the alveolar air.

In this schematic diagram, solubility in blood is represented by the relative size of the blood compartment (more soluble, larger the compartment).

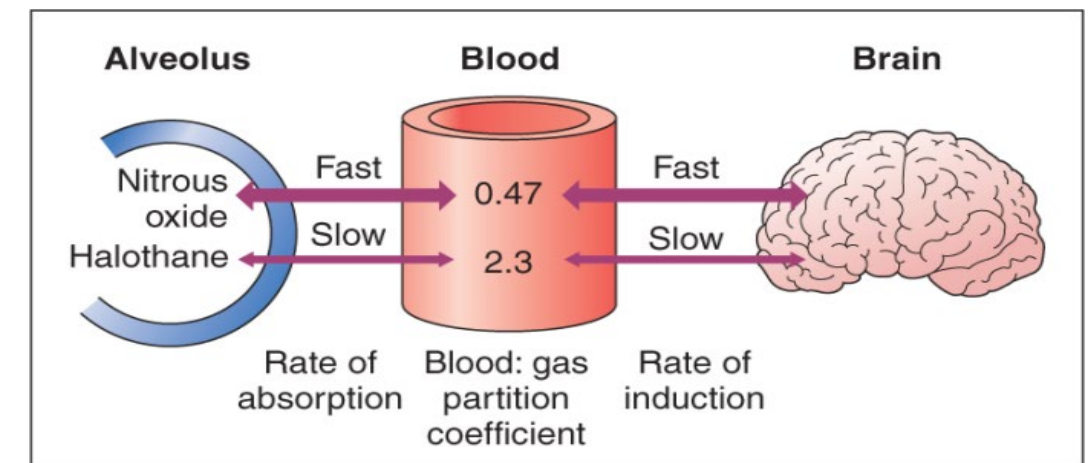
- Relative partial pressures of the agents in the compartments are indicated by the degree of filling of each compartment.
- For a given concentration of the two anesthetic gases in the inspired air, it will take much longer for the blood partial pressure of the more soluble gas (halothane) to rise to the same partial pressure as in the alveoli.

(Recall: The partial pressures in the alveoli, blood, and brain must reach equilibrium for the anesthetic to be effective.)

- Since the concentration of the anesthetic agent in the brain can rise no faster than the concentration in the blood, the onset of anesthesia will be slower with halothane than with nitrous oxide.



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Brenner & Stevens: Pharmacology, 3rd Edition.
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Clinical Key Brenner &
Stevens Pharmacology

Effects of cardiac output and alveolar-venous blood gradient on anesthesia uptake

Cardiac output relates to pulmonary blood flow:

- **↓ cardiac output** → **↓ pulmonary blood flow** → the partial pressure in blood rises more quickly → faster speed of onset (effect is more pronounced with insoluble agents)
- **↑ pulmonary blood flow** → the partial pressure in blood rises at a slower rate → the speed of onset is reduced (effect is more pronounced with soluble agents)

Alveolar-venous blood partial pressure difference and the uptake of anesthetic by tissues:

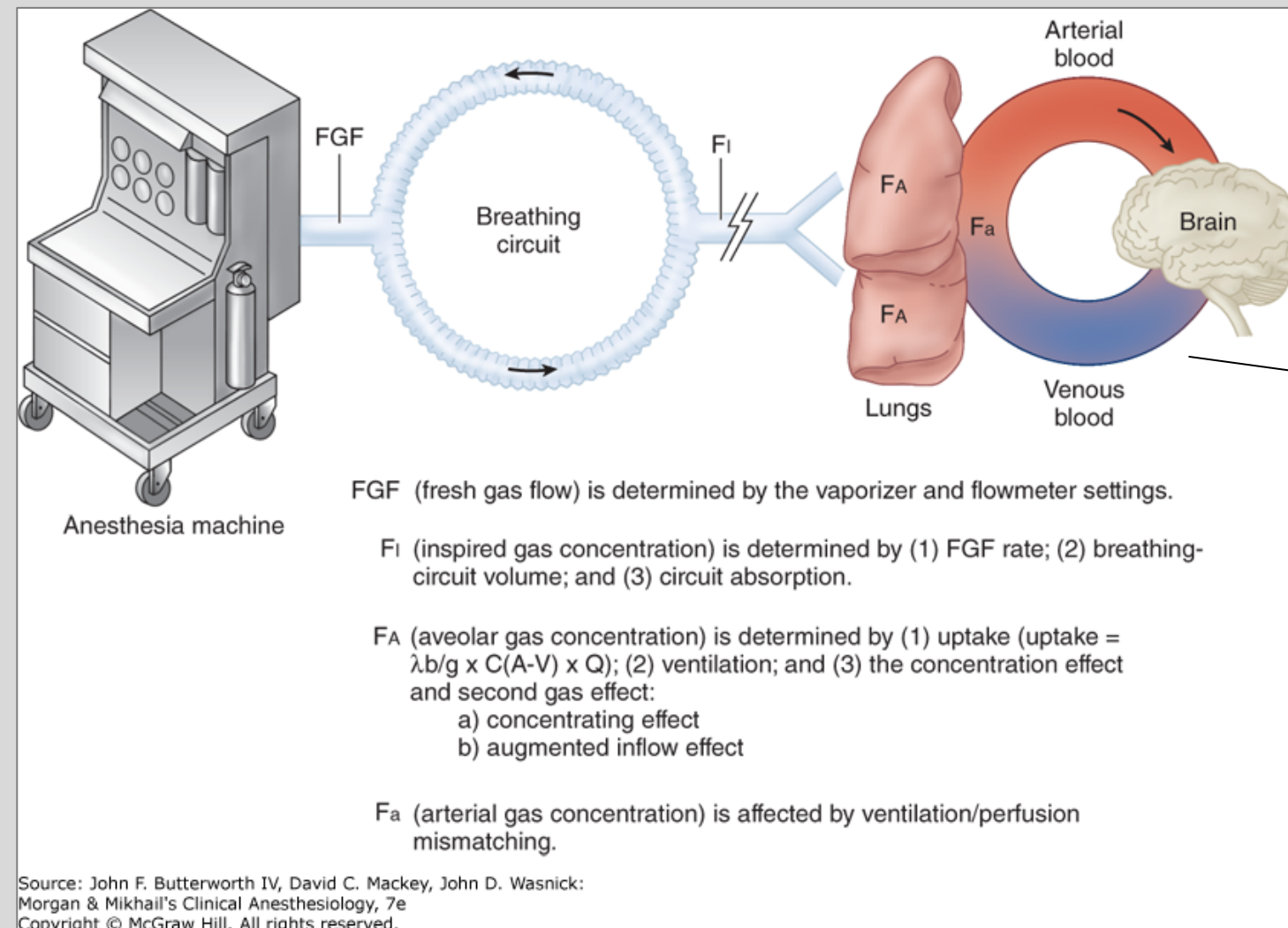
- Tissue uptake from arterial blood reduces partial pressure of the anesthetic in the venous blood.
- Venous circulation returns to pulmonary circulation with little or no dissolved anesthetic gas → generates a partial pressure difference between alveolar gas and pulmonary blood.
 - A large alveolar to venous partial pressure difference indicates peripheral gas uptake must be high.
- The greater the difference in gas concentrations, the longer the time to achieve equilibrium with brain tissue.
- Anesthetic transfer from blood to neural and non-neural tissues (tissue uptake) is determined by:
 1. Tissue solubility of the agent (blood:tissue partition coefficient)
 2. Tissue blood flow, and
 3. Difference in partial pressure gradient between arterial blood and the tissue

anesthetic gas diffuses
from arterial blood to
tissues

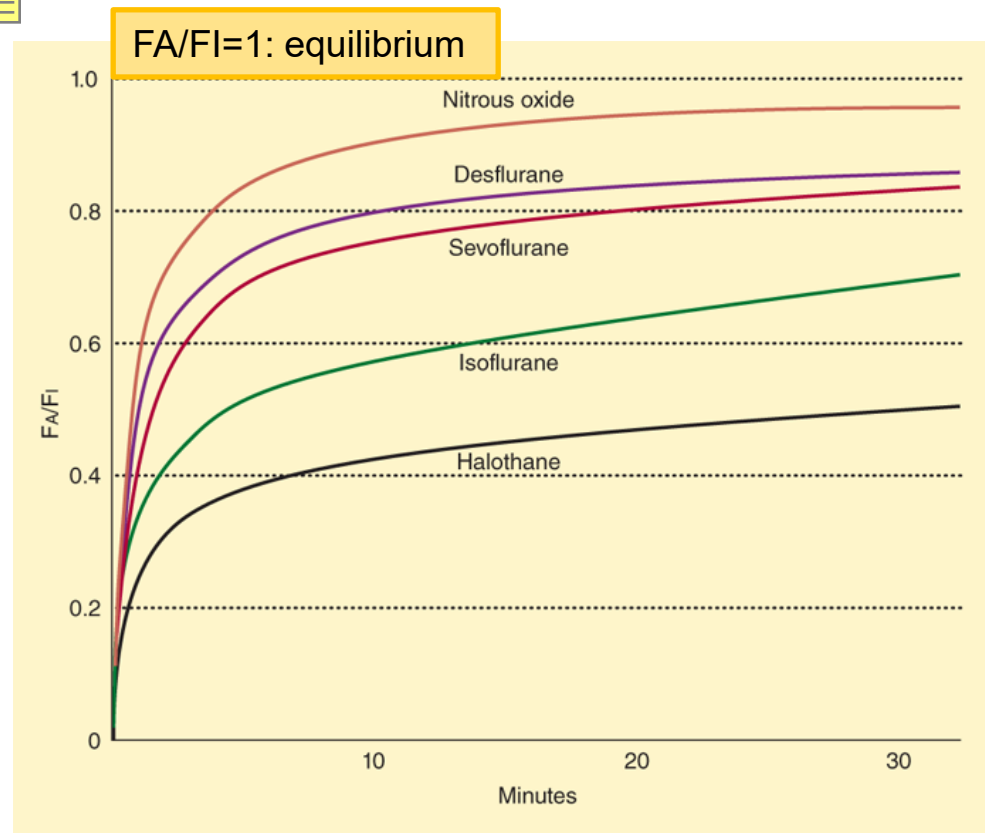
venous blood returning
to the alveoli is depleted
of anesthetic

more gas moves from
the lungs into the blood

The greater the difference in alveolar and venous gas concentrations,
the longer the time to achieve equilibrium with brain tissue.



Lower
concentration
of anesthetic in
venous blood



Source: John F. Butterworth IV, David C. Mackey, John D. Wasnick:
Morgan & Mikhail's Clinical Anesthesiology, 7e
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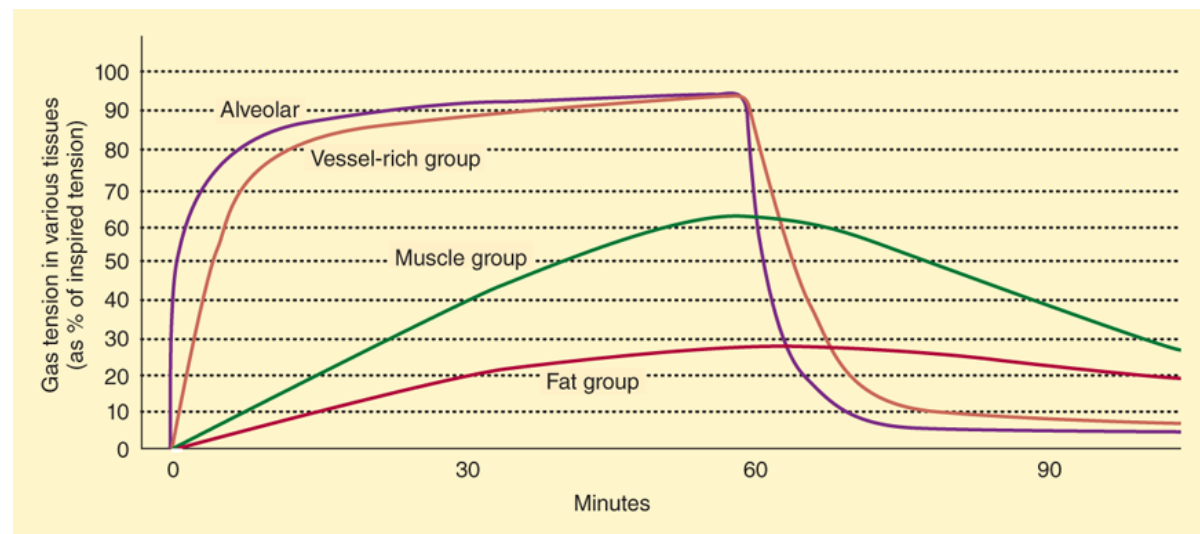
Anesthetic uptake produces a characteristic curve that relates the rise in alveolar concentration to time.

The initial steep rise in F_A/F_I is due to unopposed filling of the alveoli by ventilation.

F_A rises toward F_I faster with nitrous oxide (an insoluble agent) than with halothane (a soluble agent).

F_A : concentration of alveolar gas | F_I : concentration of inspired gas

Recall: F_A/F_I ratio is a key concept in the pharmacokinetics of inhaled anesthetic gases. It indicates how quickly the concentration of the anesthetic in the alveoli approaches the concentration of gas in the inspired air.



Source: John F. Butterworth IV, David C. Mackey, John D. Wasnick:
Morgan & Mikhail's Clinical Anesthesiology, 7e
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The shape of this graph is determined by the uptakes of individual tissue groups.

The rate of rise slows as the vessel-rich group, and eventually the muscle group, approaches steady state levels of saturation.



Tissue Groups Based on Perfusion and Solubilities

Characteristic	Vessel Rich Group	Muscle	Fat	Vessel Poor Group
Percentage of body weight	10	50	20	20
Percentage of cardiac output	75	19	6	0
Perfusion (mL/min/100 g)	75	3	3	0
Relative solubility	1	1	20	0
Concepts	First to receive anesthetic → first to approach steady state. Small volume limits capacity of uptake.	Less perfused so uptake is slower. Greater capacity for uptake due to greater tissue volume → uptake sustained for hours	Perfusion similar to muscle group but tremendous solubility of anesthetic in fat increases total capacity → days to approach steady state	Minimal perfusion → insignificant uptake
Vessel rich group: highly perfused tissue: brain, heart, liver, kidney, endocrine organs Vessel poor group: minimally perfused tissue: bones, ligaments, teeth, hair, cartilage Steady state: arterial and tissue partial pressures are equal				

Adapted from Access Medicine Morgan & Mikhail's Clinical Anesthesiology, 2022, Table 8-2

Check your knowledge

What is the fundamental concept that ensures that the anesthetic concentration in the CNS is stable and effective for maintaining anesthesia during the procedure?

What four factors influence the uptake of anesthesia?

Why does faster the uptake of anesthetic agent lead to a slower speed of induction?

Check your knowledge

Why is the speed of induction slower with the more soluble anesthetic gases?

How does an increase in cardiac output lead to a slower speed of induction?

How does a large alveolar to venous partial pressure difference influence the speed of induction?

Recovery and Elimination of inhaled anesthetics

Recovery: Anesthetic transfer in reverse

Elimination

Solubility: Lower the solubility → faster the elimination	Lungs: Exhalation is the major route.
Ventilation: ↑ ventilation → ↑ rate of elimination	Liver: some drugs
Exposure: Longer exposure → greater tissue saturation with drug → prolonged elimination time	Rank order of drugs eliminated by hepatic metabolism: Methoxyflurane > halothane > enflurane > sevoflurane > isoflurane > desflurane > N ₂ O ~5% (CYP2E1) <0.2% 0.02% <0.004% (These values are listed to provide perspective.)
Anesthetic circuit volume: The lower the anesthetic circuit volume, the faster the elimination of inhaled anesthetic.	

Methoxyflurane, halothane, and enflurane are no longer used in U.S.

Minimum Alveolar Concentration (MAC)

MAC is a standardized measure of **potency** and allows for consistent **dosing** across different patients and clinical settings.

- 1 MAC = the alveolar concentration that prevents movement in 50% of patients in response to a standardized stimulus (eg, pain of surgical incision)
- MAC should be considered ***a statistical average (median)***.
- MAC can be altered by several physiological and pharmacological variables: age, pregnancy, acidosis, IV anesthetics, opioids, and more.

Potency

MAC is expressed in vol%:

1 MAC of N₂O = 105 vol%

1 MAC Halothane = 0.75 vol%

1 MAC Sevoflurane = 2 vol%

1 MAC Desflurane = 6%

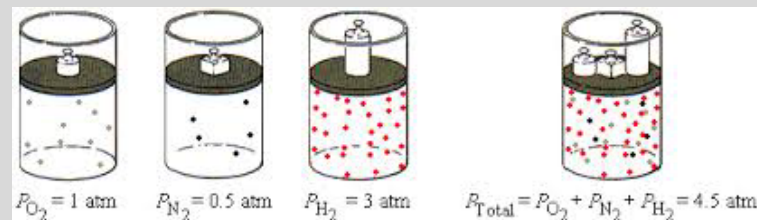
Dosing

Usual dose is 0.5-1.5 MAC

MAC values are additive:

0.5 MAC N₂O + 0.5 MAC isoflurane
= 1 MAC of anesthetic gas

vol %: the percentage of vapor pressure at 1 atmosphere that is the anesthetic



From: **13 Anesthetics**

Lippincott® Illustrated Reviews:
Pharmacology, 7e, 2019

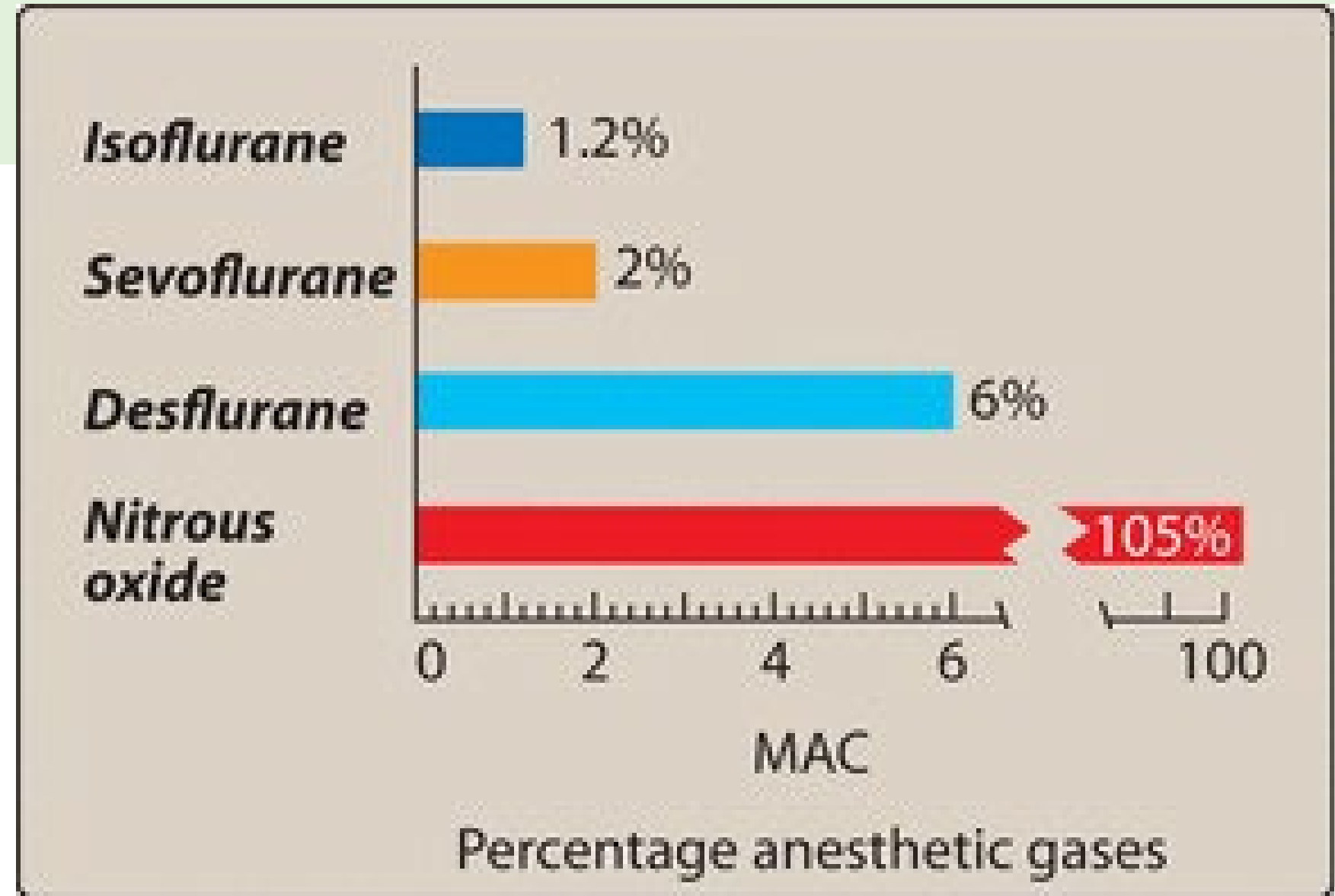
Clinical Utility of MAC

- **Correlation:**

MAC provides a direct correlation between the alveolar anesthetic concentration and the concentration at the sites of action in the CNS.

- **Comparison:**

MAC permits comparison of the relative anesthetic potencies of the different anesthetics.



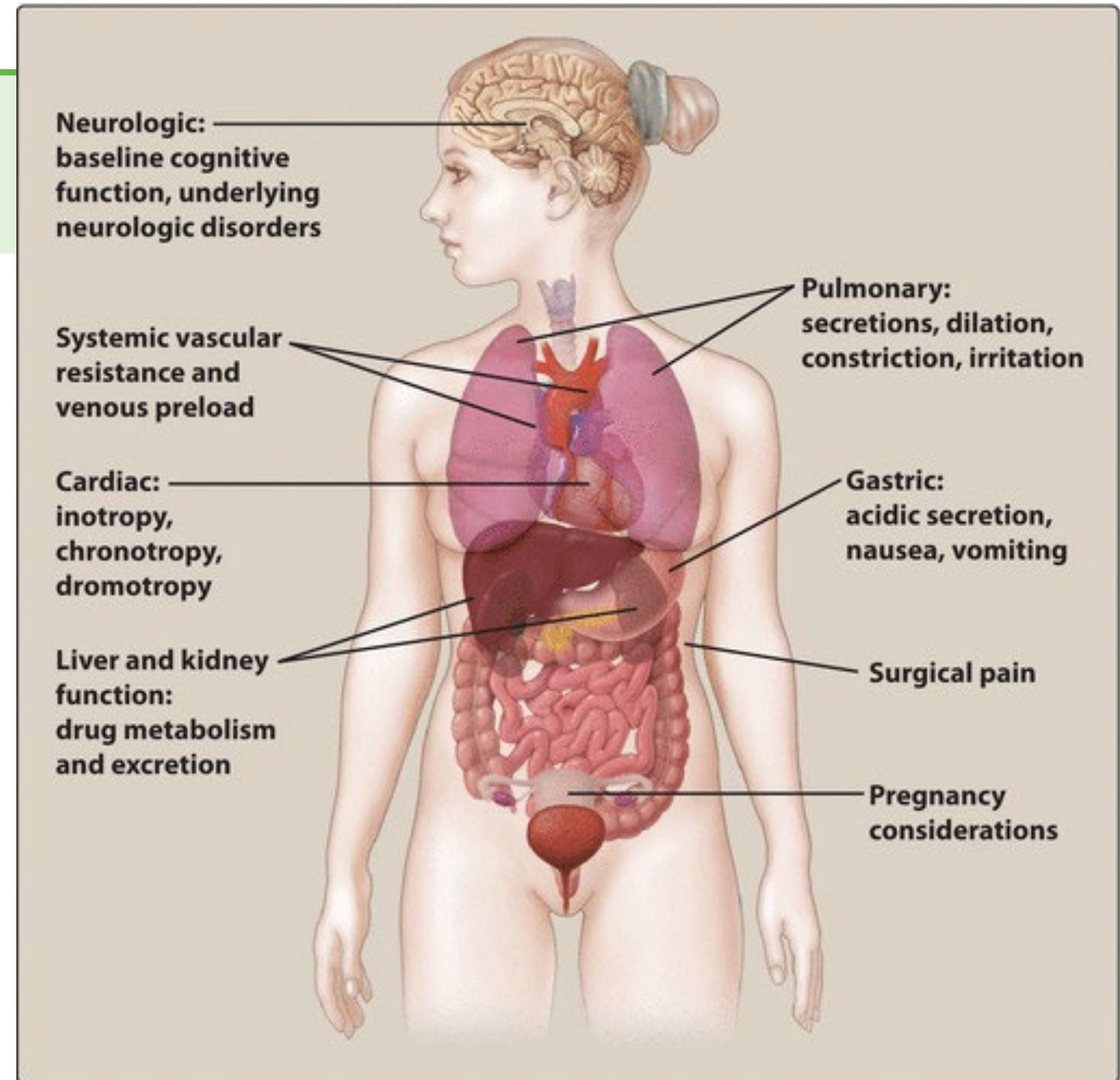
Minimal alveolar concentrations (MAC) for anesthetic gases are used to compare pharmacologic effects of different agents (high MAC = low potency).

From: **13 Anesthetics**

Lippincott® Illustrated Reviews:
Pharmacology, 7e, 2019

Overall patient considerations when administering an anesthetic.

- Patient history: Previous reactions to anesthetic, allergies, medication
- Age and weight
- Cardiovascular health
- Respiratory function
- Liver and kidney function
- Neurological status, especially seizure history
- Aspiration risk
- Pregnancy
- Drug interactions
- Postop care and monitoring: pain management, nausea/vomiting, delirium
- Patient consent and education



Summary

Anesthetic	B:G	B:B	¹ MAC	Comments
Nitrous oxide	0.47	1.1	>100	Weak anesthetic; rapid onset and recovery Not metabolized Can cause expansion of air-fill cavities / Diffusional hypoxia Prolonged exposure can result in megaloblastic anemia, peripheral neuropathies
Desflurane	0.42	1.3	6-7	Low volatility; rapid onset/recovery; poor induction agent (respiratory irritation) SNS stimulation → transient, often marked, tachycardia, ↑BP Carbon monoxide can be produced from desiccated CO ₂ absorbents in breathing circuit
Sevoflurane	0.69	1.7	2.0	Rapid onset/ recovery Smooth rapid inhalation induction; bronchodilating – preferred for patients with airway problems Hepatic metabolism (~5% of absorbed drug) Interaction with soda-lime → Compound A vinyl ether (renal toxicity not reported)
Isoflurane	1.40	2.6	1.4	Medium rate of onset and recovery; pungent odor → poor induction agent; dilates coronary arteries
Enflurane	1.80	1.4	1.7	Medium rate of onset / recovery; limited biodegradation; seizure potential
Halothane	2.30	2.9	0.7	Relatively slow onset / recovery; significant hepatic metabolism; halothane hepatitis – reactive intermediary – direct toxicity, immune reaction

¹MAC=median alveolar concentration in vol%: the anesthetic concentration that produces immobility in 50% of patients exposed to noxious stimulus; MAC is small for potent anesthetics and large for less potent anesthetics.

B:G=blood:gas partitions coefficients; B:B=brain:blood partition coefficients

Adapted from Basic & Clinical Pharmacology Table 25-2



Common advantages and disadvantages of potent volatile anesthetics

Advantages

- Ease of administration
- Reliable block of sensory input (ie, pain)
- Bronchodilation
- Decrease in skeletal and smooth muscle tone (dose-dependent)
- Decreased cerebral metabolic rate
- Increased cerebral blood flow by direct vasodilatory effect
- Monitoring end-tidal anesthesia concentration as a measure of anesthetic depth

Disadvantage

- Dose-dependent...
 - Suppression of airway reflexes
 - Respiratory depression
 - Hypotension due to myocardial depression and vasodilation
- Increased post-op nausea/vomiting compared to IV anesthetics (antiemetic needed)
- Malignant hyperthermia potential in susceptible individuals

Nitrous Oxide: Summary of effects

Anesthetic effects	Incomplete (weak) anesthesia ➤ Adjunctive use with inhalation and intravenous anesthetics for sedation, analgesia, and anxiolytic effects
Muscle effects	Does not relax skeletal muscle → does not enhance the effects of neuromuscular blocking drugs Does not relax uterine smooth muscle ➤ Does not cause malignant hyperthermia
Organ effects	Sympathetic stimulation → maintains hemodynamics Increases respiratory rate Reduces hepatic and renal blood flow not hepato- or nephrotoxic
Other effects	• Body space volume expansion and increased pressure in spaces with noncompliant walls. • Interferes with vitamin B12 function; avoid in patients with disorders of B12 deficiency and folate or methionine synthesis • Diffusion hypoxia • Second gas effect (concentration effect)

Check your knowledge

What is the fundamental concept that ensures that the anesthetic concentration in the CNS is stable and effective for maintaining anesthesia during the procedure?

- When the partial pressure of the anesthetic gas in the inspired air, alveoli, arterial blood, and CNS are in equilibrium (steady state, $F_A/F_I=1$), the desired clinical effect will be produced.

What four factors influence the uptake of anesthesia?

- Inspired concentration of anesthetic gas
- Solubility of gas in blood (defined by the blood:gas partition coefficient)
- Cardiac output
- Alveolar-venous partial pressure difference

Why does faster the uptake of anesthetic agent lead to a slower speed of induction?

- When uptake is fast, a large amount of anesthetic is absorbed into the blood from the alveoli → creates a significant difference between the inspired concentration and the alveolar concentration ($F_A < F_I$) as the alveoli are depleted of anesthetic faster than they can be replenished by inhalation → it takes longer to reach equilibrium ($F_A/F_I=1$)

Check your knowledge

Why is the speed of induction slower with the more soluble anesthetic gases?

- The more soluble the anesthetic is in blood (the higher the partition coefficient) → the greater the capacity for uptake into the blood → the slower rate of rise in partial pressure → the longer it takes to reach equilibrium

How does an increase in cardiac output lead to a slower speed of induction?

- ↑ cardiac output → ↑ pulmonary blood flow → the partial pressure of the gas in blood rises more slowly → slower rise in F_A/F_I towards equilibrium → slower speed of onset
- This effect is more pronounced with soluble agents, which achieve equilibrium slower.

How does a large alveolar to venous partial pressure difference influence the speed of induction?

- Anesthetic gas is taken up from arterial blood into nonneural tissues → venous circulation returns to pulmonary circulation with little or no dissolved anesthetic gas → generates a partial pressure difference between alveolar gas and pulmonary blood.
- The greater the difference in gas concentrations, the longer the time to achieve equilibrium with brain tissue.

Check your knowledge

What is the acronym MAC stand for?

- Minimum alveolar concentration

How does MAC define potency?

- 1 MAC = the concentration of the inhaled gas in the alveoli that prevents movement in 50% of patients in response to a standardized stimulus (eg, surgical incision) at standard temperature and ambient pressure.
- MAC reflects brain partial pressure and allows comparisons of potency between agents.

How is MAC used in dosing inhaled anesthetics?

- Dosing of inhaled anesthetics is guided by MAC and MAC values are additive.

Which inhaled anesthetics are potent volatile agents in common use?

- Isoflurane, desflurane, sevoflurane

Why is nitrous oxide referred to as a weak anesthetic?

- Nitrous oxide has low potency. Even 100% nitrous oxide does not achieve 1 MAC. Therefore, it is combined with a potent volatile anesthetic.

Check your knowledge

Why may isoflurane, desflurane, and sevoflurane be suitable for patients with impaired myocardial function?

- Isoflurane, desflurane, and sevoflurane cause vasodilation, which lowers systemic vascular resistance and blood pressure with minimal change in cardiac output (preserves cardiac output) – of course, with close monitoring to ensure patient safety and optimal outcomes.

What is the concern with the use of desflurane in patients with coronary artery disease?

- Although desflurane causes vasodilation with minimal effect on the force of contraction (inotropy), its transient sympathomimetic properties cause tachycardia and hypertension, especially at high dose. These cardiac effect may induce cardiac ischemia and heart attack.

Which inhaled anesthetics are associated with malignant hyperthermia in patients with a genetic disorder in skeletal muscle calcium regulation?

- The halogenated anesthetics: isoflurane, desflurane, sevoflurane, halothane, enflurane
- Not nitrous oxide.

Check your knowledge

Match the toxicities with the drugs that cause them

Body space volume expansion	Nitrous oxide
Carbon monoxide production in desiccated CO2 absorber	Halogenated volatile anesthetics
Cortisol synthesis inhibition	Isoflurane
Diffusion hypoxia	Nitrous oxide
Environmental toxicity	All inhaled anesthetics
Malignant hyperthermia	Halogenated volatile anesthetics
Myoclonus	Isoflurane
Nausea and vomiting	All inhaled anesthetics
Nephrotoxicity (potentially) by compound A	Sevoflurane
Sympathetic stimulation → tachycardia and hypertension	Desflurane
Vitamin B12 inhibition	Nitrous oxide

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- Lexidrug monographs
- UpToDate various articles; literature current through Dec 2024

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